



Arab American University Palestine

**Faculty of Engineering and Information
Technology**

Rawasi: Palestinian Cultural Shop

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List of abbreviation

ERD	Entity Relation Diagram
NLP	Natural Language Processing
GAI	Generative Artificial Intelligence
UI	User Interface
AI WIKI	Artificial Intelligence Wiki
API	Application Programming Interface
LLM	Large Language Model
UML	Unified Model Language

Declaration

This is to declare that the graduation project entitled (**Rawasi**) under the supervision of Dr. Amjad Rattroot is our own work and does not contain any unacknowledged work or material previously published or written by another person, except where due reference is made in the text of the report.

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Date: June 2026

Introduction

Like many young people, we learn about our culture and heritage through the elderly in our families. But this isn't possible for many people nowadays. And trying to learn online can sometimes be confusing and time-consuming.

Preserving our culture is crucial. Our generation has to use huge advancements in technology to make that happen. As technology students ourselves, we feel it's our duty to make use of our skills to preserve our culture.

We want to keep the Palestinian heritage alive in young people's hearts and minds, but not only that! We aim to make it fun to receive, easy to access, and all in one place. And here's where our project comes to the rescue.

The proposed project is a smart application with a heartwarming theme: a traditional store. It is supported by a multi-agent Chatbot platform. When you enter our shop, you will find its owners greeting you, ready to answer any questions you have.

Proposal

(**Rawasi**) is a 3D game-like application that looks like a traditional Palestinian shop selling traditional goods. The user can roam around and explore all the items in the shop. And if any confusion comes up, or any question comes to mind, the agents in the app will do the explaining.

We introduce the two main AI features; the chatbots. Two agents represent the owners of the shop. The first is an old man who has knowledge of the goods in store that you can explore, such as clothing, instruments, soups and oils, and so on. The second one is his wife, an old lady who knows all about the intangible parts of the culture. She talks about Palestinian tales, proverbs, songs, and more!

Note: the terms **Um Khalil/ Abu Khalil**, Grandfather/Grandmother, Artifice Keeper/Storyteller. Will be interchangeably used throughout the document; they all refer to the same two characters! Our lovely shop owners!

Chapter 2: Feasible Study

2.1 Overview

This chapter presents and discusses existing systems, websites, and applications that are related to our project idea, such as digital platforms for Palestinian heritage, virtual museums, digital archives, and interactive cultural applications. The aim of this chapter is to study these systems in terms of content presentation, level of interaction, use of modern technologies, and user experience. It also highlights the differences between these systems and our application, which focuses on presenting Palestinian heritage through an interactive 3D virtual shop, rotatable 3D objects, and chatbot characters.

2.2 Existing Systems

1.The Palestinian Museum Website: is a digital platform that presents Palestinian culture and history, educational programs, research, and digital content [1].

It also includes the “Museum from Home” section, which allows users to explore cultural and educational content remotely. The website is also connected to a digital archive that helps preserve Palestinian memory through historical photos, documents, and cultural materials.

Compared with our application, the Palestinian Museum Website focuses more on browsing, reading, and presenting information. Our application is more interactive because it allows users to explore a 3D heritage shop, tap on heritage items, view them as rotatable 3D models, and interact with chatbot characters such as grandfather and grandmother.

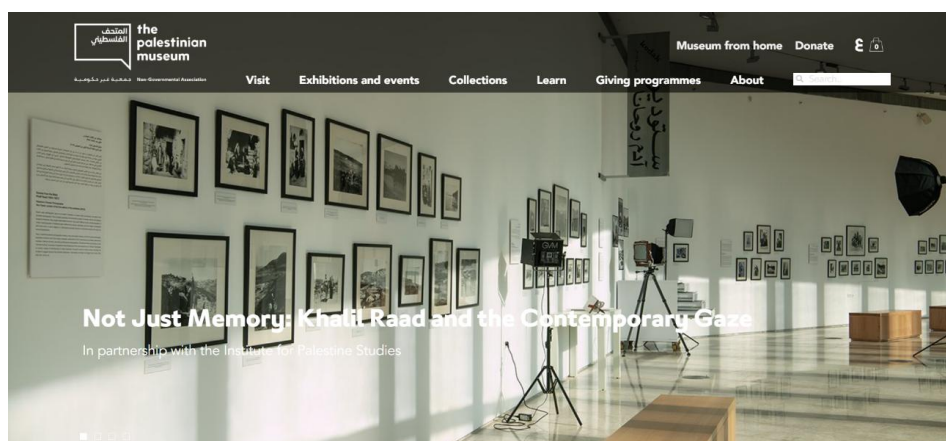


Figure 2.1 1

2. Google Arts & Culture

Is a digital platform and mobile application by Google that allows users to explore artworks, museums, cultural objects, and historical places from around the world. It also offers digital exhibitions, high-quality images, stories, and some virtual tours.

Compared to our application, Google Arts & Culture is broader and more global. It focuses on browsing artworks and cultural collections, but our application focuses specifically on Palestinian heritage through an interactive 3D shop, rotatable 3D objects, and chatbot characters such as the grandfather and grandmother [2].

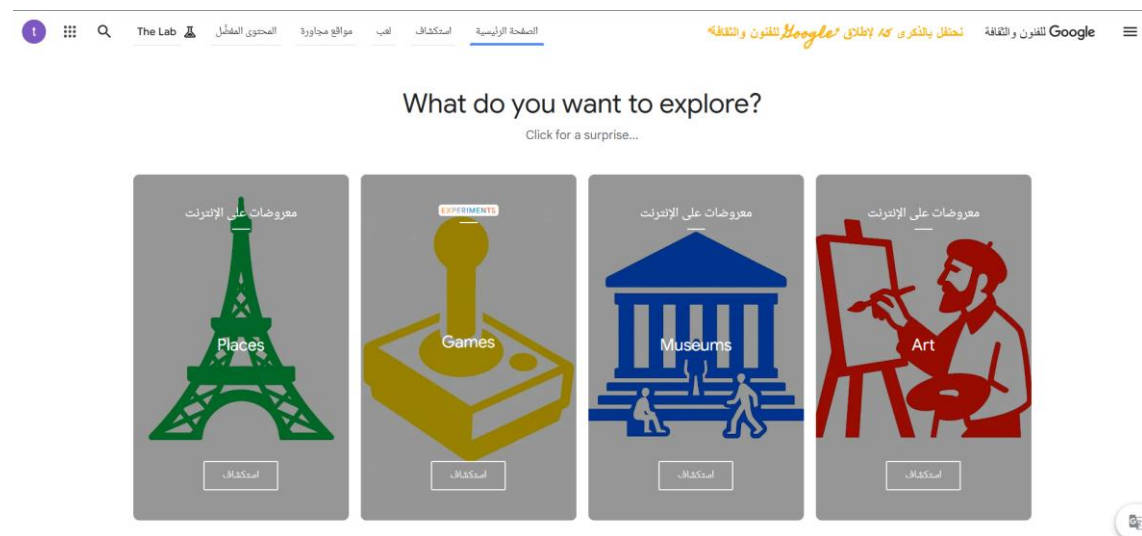


Figure 2.1 2

3. The Palestinian Heritage Trail:

is a long cultural walking trail in Palestine that aims to introduce visitors to Palestinian nature, villages, cities, historical sites, and hospitality. The trail extends for hundreds of kilometers and passes through different Palestinian areas, combining tourism, nature, and local heritage exploration [3].

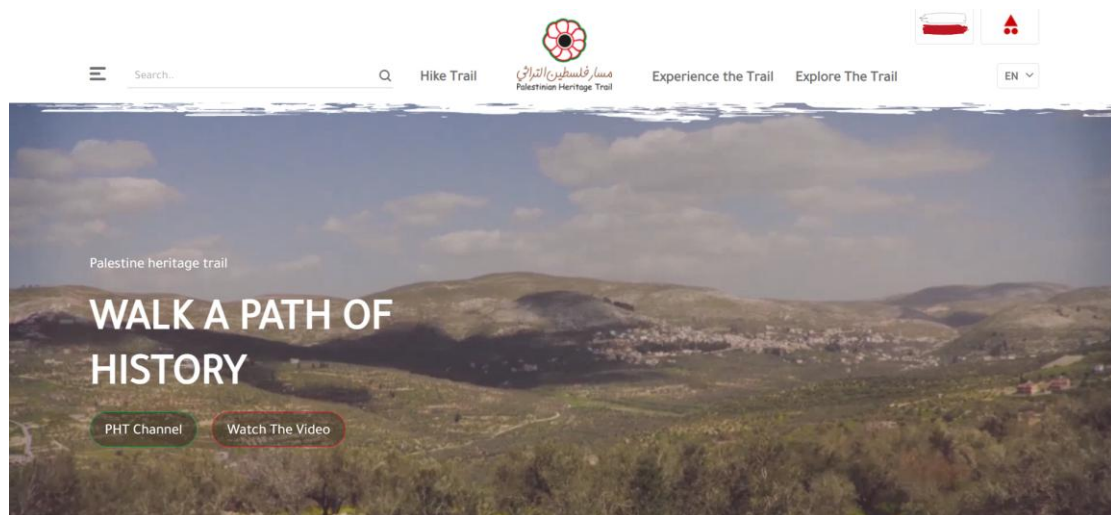


Figure 2.1 3

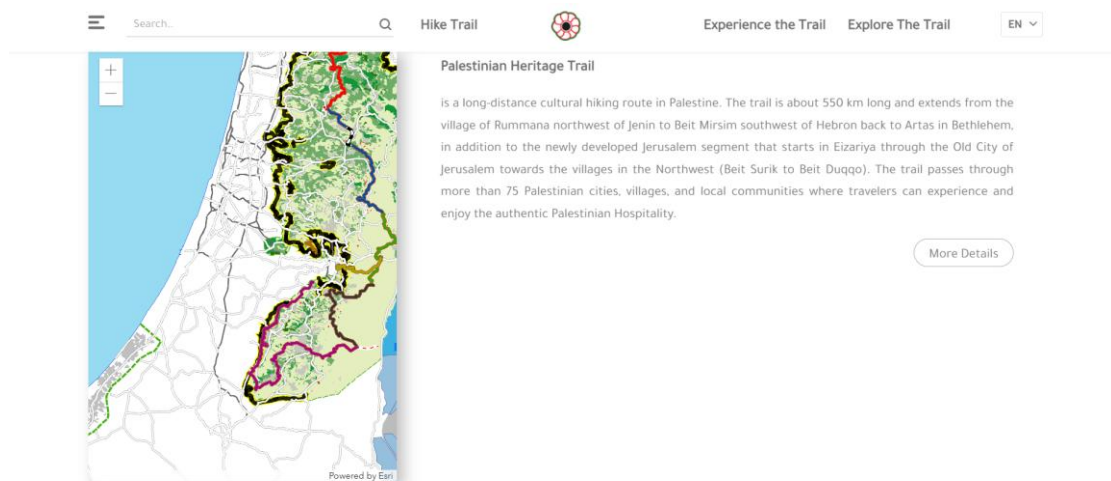


Figure 2.1 4

4. CYARK:

is a website that focuses on documenting and preserving cultural heritage sites around the world using 3D models. The website provides three-dimensional virtual tours of historical and cultural sites, allowing users to explore heritage places digitally through images, 3D models, and interactive storytelling. It aims to make global heritage accessible to the public and future generations by using 3D technology for digital documentation and preservation [4].

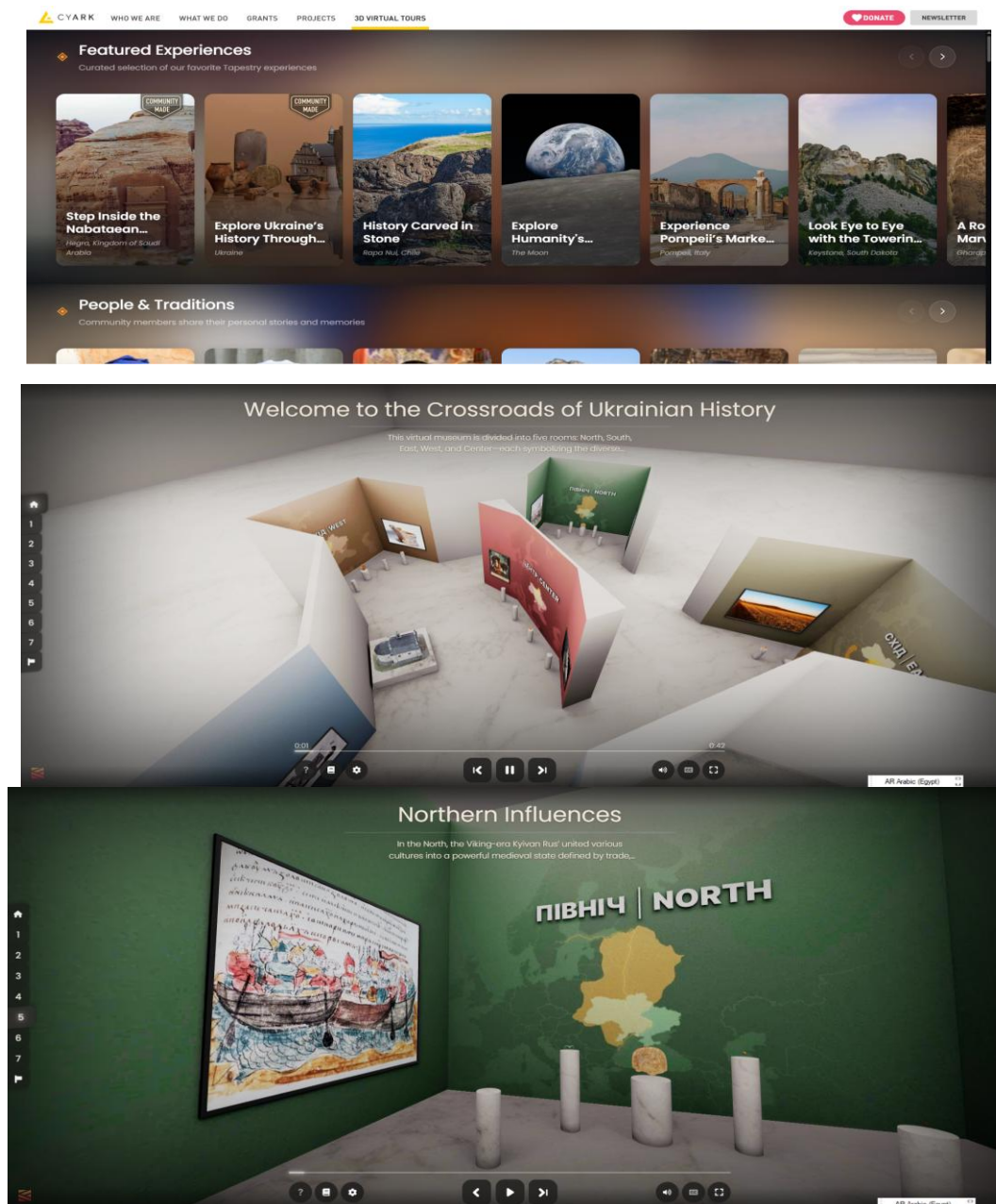


Figure 2.1 5

Feature	CYARK	Google Arts & Culture	Palestinian Heritage Trail	Palestinian Museum Website	Rawasi Application
Presents Palestinian heritage	No, global heritage	Partially	Yes	Yes	Yes
Provides historical information	Yes	Yes	Yes	Yes	Yes
Educational content	Yes	Yes	Yes	Yes	Yes
Real-life tourism / walking trail	No	No	Yes	No	No
Shows real places	Yes	Yes	Yes	Yes	No, virtual shop
3D environment	Yes	Sometimes	No	No	Yes
Virtual tour	Yes	Yes	No	Limited	Yes
Virtual heritage shop	No	No	No	No	Yes
Object interaction	Limited	Limited	No	Limited	Yes
Rotatable 3D objects	Yes	Sometimes	No	No	Yes
Chatbot characters	No	No	No	No	Yes
Storytelling through characters	No	Limited	No	No	Yes
Focus on stories and proverbs	Limited	Limited	Limited	Limited	Yes
Mobile application experience	No	Yes	No	No	Yes
Suitable for children and young users	Moderate	Moderate	Moderate	Moderate	Yes
Focus on interactive learning	Moderate	Moderate	Limited	Limited	Yes

Table 2.11

2.3 Summary:

Based on the reviewed systems, several platforms and websites aim to preserve and present Palestinian heritage and cultural content, such as the Palestinian Museum, the Palestinian Heritage Trail, Google Arts & Culture, and CYARK . These systems provide important information, images, documents, and cultural resources that help preserve Palestinian memory and identity. However, most of them mainly rely on browsing, reading, or displaying images and documents, and they do not provide a fully interactive 3D learning experience.

Our application is different because it presents Palestinian heritage in a more interactive way. Users can enter a virtual heritage shop, navigate inside it, tap on heritage objects, view them as rotatable 3D models, and interact with chatbot characters such as grandfather and grandmother. Therefore, the application not only displays heritage but also transforms it into an engaging educational experience suitable for children, young users, and the general audience.

Chapter 3: Theoretical background

Stages of developing a game

3.1 Pre-Production (Planning & Design)

3.1.1 Concept & Pitch:

Genres: Educational

Core Idea:

An educational game-like application in the theme of a Palestinian cultural shop (might look like the shops in the old Palestinian markets), users can explore and navigate through the 3D environment, the shop will be a one big room where each corner is a section that displays a category for Palestinian traditional crafts. The owners of this shop will be the user's guide! Two lovely characters who will be the chatbots that are always present to answer any question the user has.

3.1.2 Game Design

Story:

Users role-play as tourists who visit a Palestinian cultural shop in Jerusalem, a regular visitor will be always remembered and loved by the shop's owners.

Characters:

Grandfather(ابو خليل) : is a Palestinian man in his 70s who traveled a lot across Palestine and visited many places, interacting with different people from different areas. The information and experiences he gained from his travels allowed him to become very knowledgeable about the tangible parts of the heritage.

Grandmother (ام خليل): is a lovely lady who has lived a beautiful life and also interacted with many people throughout Palestine. She is a very fun person to talk to and is deeply loyal to her culture. She enjoys narrating all types of stories and often uses proverbs during her narration, making conversations with her feel lively and engaging.

Um Khalil and **Abu Khalil** are both married; they had their shop “Rawasi” for as long as they remember! Pouring their hard work and passion into this lovely place.

Gameplay:

Players enter the cultural shop, where they can interact with the 3D objects by clicking on them. They can also interact with the two characters in the shop by asking them questions through the chatbot interface.

When asking the first character, the grandfather, players can get information related to the tangible part of the heritage. This character can answer questions about clothing, instruments, trees, food, and anything gathered from his travels and experiences.

When interacting with the second character, players can hear stories, songs, traditions, and proverbs that represent the intangible side of the heritage. While interacting with both characters, players can also experience how each character behaves. Both give off warm vibes, but each has a certain type of speech and personality that makes the experience feel more interactive, as if the player were talking to an actual older person who is close to them.

In addition to the conversations, players can closely interact with the 3D cultural objects displayed in the shop. The objects can be viewed from all angles, allowing users to explore every detail up close. For example, when clicking on a traditional dress, the dress becomes the main focus on the screen while the background is blurred. The 3D object is then displayed directly in front of the player, who can rotate it and inspect all of its details. This allows the user to have a more immersive and enjoyable exploration experience.

Mechanics: interacting with the 3D objects and talking to the two AI agents

3.1.3 UI design:

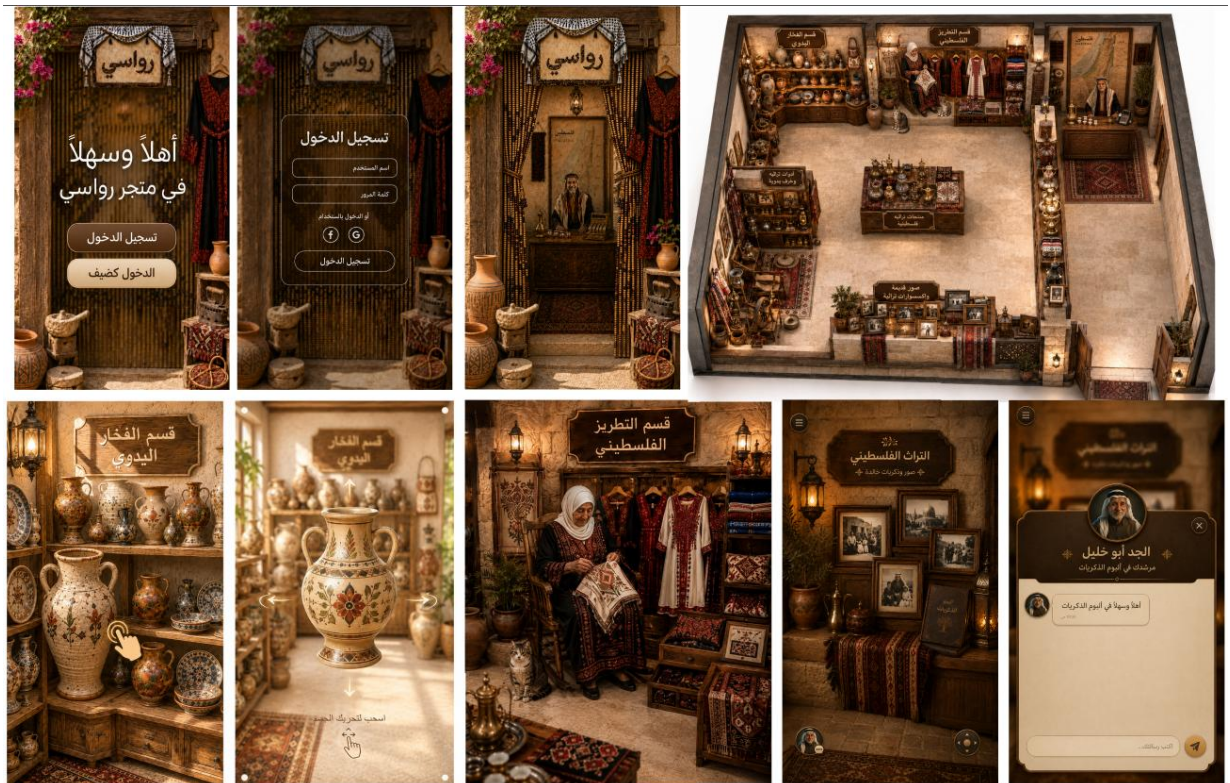


Figure 3.1 1

3.2 Production (creating assets/coding)

3.2.1 Features and specifications

External users

Specifications:

1. External users and only access to information allowed by the system.
2. They can interact with information via chatbots and receiving answers.
3. but they haven't permission to delete or modify the information.

Features:

1. Search and explore information via chatbots.
2. Ask questions
3. View stories and access to artifacts

Artifact Keeper agent

Specifications

1. The Artifact **Keeper agent** manages artifacts such as documents, summaries, generated outputs, and media files.
2. It matches artifacts to the correct topic in AI wiki to give information about artifacts.

Features:

1. Save artifacts
2. Organize artifacts
3. Retrieve information

Storyteller agent

Specifications:

1. The Storyteller **agent** uses information stored inside the AI wiki
2. connects the artefacts to suitable story or intangible information in simple and cultural tone.

Features:

1. Generate stories
2. Summarize information
3. Explain artifacts and match cultural artifacts.
4. Convert technical content into simple language.

AI wiki

Specifications:

1. AI wiki is a main knowledge base of the platform.
2. it stores structured and a structured data.
3. Users can search about content via chatbots.

Features:

1. Store information
2. Search Information
3. Retrieve related contents
4. Link between wiki topics and artifacts

Wiki admin

Specification:

1. The AI wiki admin controls and maintains the AI wiki.
2. The admin ensures that the information is accurate, last updated, and organized.
3. the admin can also control who can access and edit specific contents.

Features:

1. Add, modify and delete contents
2. Manage user permissions
3. Review and approve contents
4. Monitor wiki quickly

3.2.2 Existing technologies that will be used in Rawasi

Chatbots:

Two main chatbots will be programmed for this project. The first chatbot is a grandfather who's the shop owner that will guidelines tourists(users) and give them accurate information about the items he sells.

The second chatbot is grandmother who specialist in intangible information about Palestinian heritage that interact with user and make relation between objects and intangible information.

NLP:

Natural language processing inside our project we use NLP that ensure the text that user written it is a suitable structure, and generative AI can analyze it correctly to ensure full accurate response and understandable by user.

AI wiki / Wikimedia:

Is a reference that stores tangible and intangible heritage information used to interact with the user and give him a good answer about his question via chatbots or clicked on object inside the environment.

Generative AI:

Is a tool that accepts text(prompt) via chatbots and generate full structured and understandable information in Palestinian culture writing mood.

The development team of Rawasi will most likely be using Gemini as a Generative AI tool because it is a low-cost token and easy to integrate into this project.

3.2.3 Existing Tools

Layer	Technology	Reason for using this tool
Mobile 3D game app	Unity + C#	Unity is very suitable for mobile 3D games. It supports Android and iOS builds, 3D environments, object clicking/tapping, zooming, rotating, animations, avatars, and game-like navigation.
Mobile UI inside game	Unity UI / TextMeshPro	Use this for the in-game chatbot window, buttons, menus, object information panels, Arabic text, and popup messages.
Authenticatin	Firebase Authentication	Provides ready-made login and registration for mobile users without building the full authentication system manually.
Main database	Firestore	Stores users, artifacts, stories, songs, proverbs, chat messages, interaction logs, and admin-approved AI Wiki content. It also works well with real-time updates.
File storage	Firebase Storage	Stores 3D models, images, PDFs, cultural documents, audio files, and other media used inside the mobile game.
Automation workflows	n8n	Handles chatbot workflows, routes user questions to the correct agent, processes admin uploads, sends alerts, and connects Firebase with AI services.
AI agents	n8n + OpenAI / LLM API	n8n organizes the Grandfather and Grandmother agent workflows, while the LLM API generates natural cultural answers.
Backend logic	Firebase Cloud Functions	Best fit with Firebase for mobile. It can securely connect the mobile app to n8n, process chatbot requests, validate data, and trigger AI workflows.
Admin panel	React + TypeScript	The admin panel is still useful as a web dashboard for adding cultural data, uploading files, reviewing chatbot answers, and correcting AI Wiki information.

Admin hosting	Firestore Hosting	Hosts only the admin dashboard, not the mobile game.
Mobile deployment	Android APK / AAB and iOS build from Unity	Unity can export the game as an Android APK/AAB and iOS project, so users install it like a normal mobile game.

Table 3.1 1

3.2.4 Interaction Flow

Interaction Object Flow

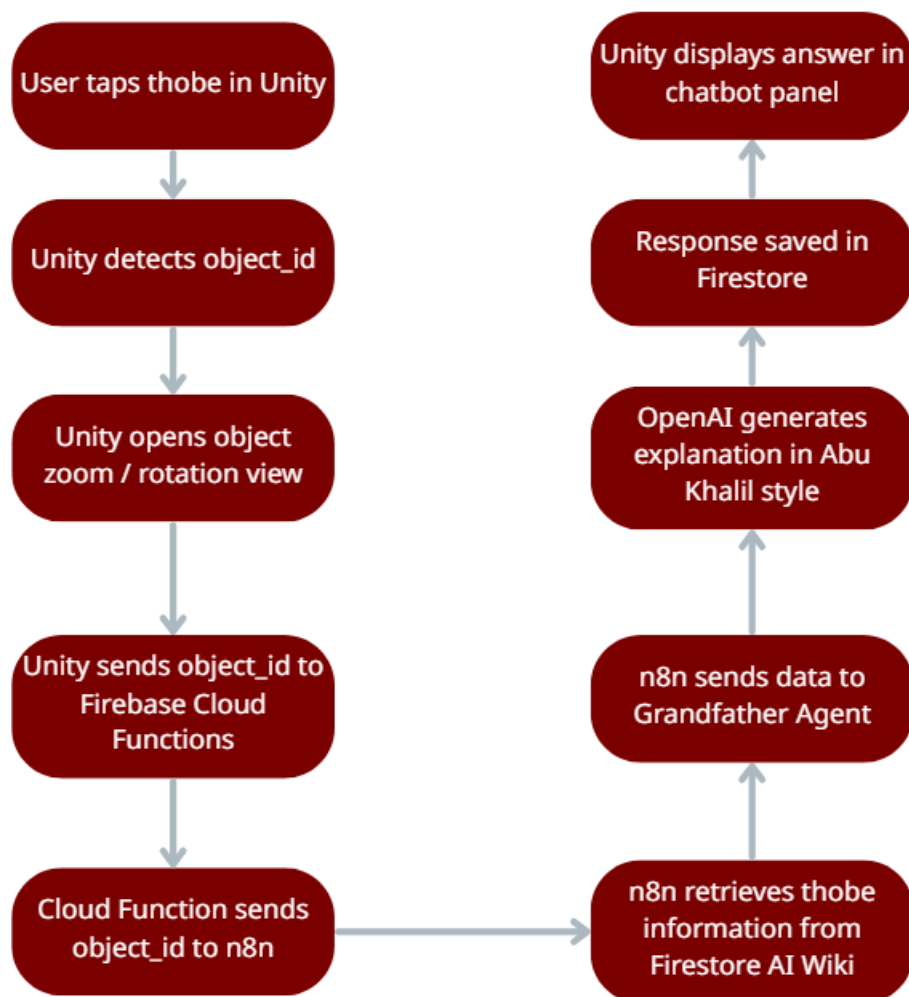


Figure 3.1 2

Chatbot Question Flow

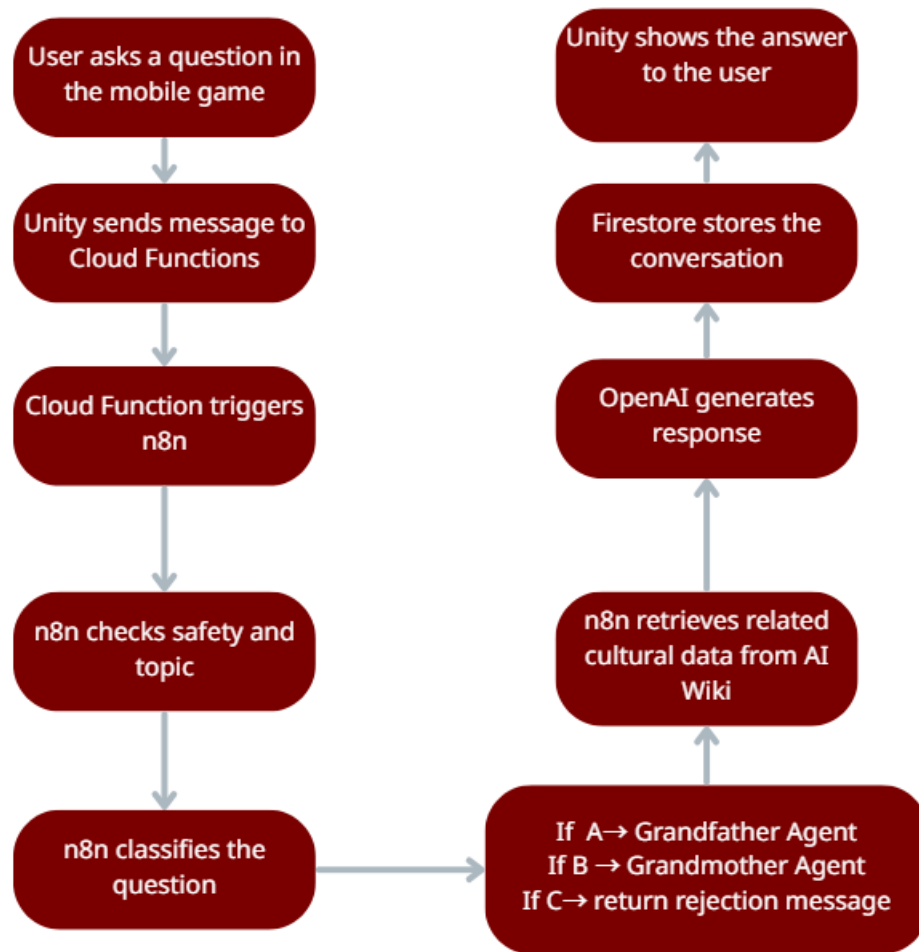


Figure 3.1 3

A: object/artifact/food/tool

B: story/song/proverb/tradition

C: unrelated

Admin Update Flow

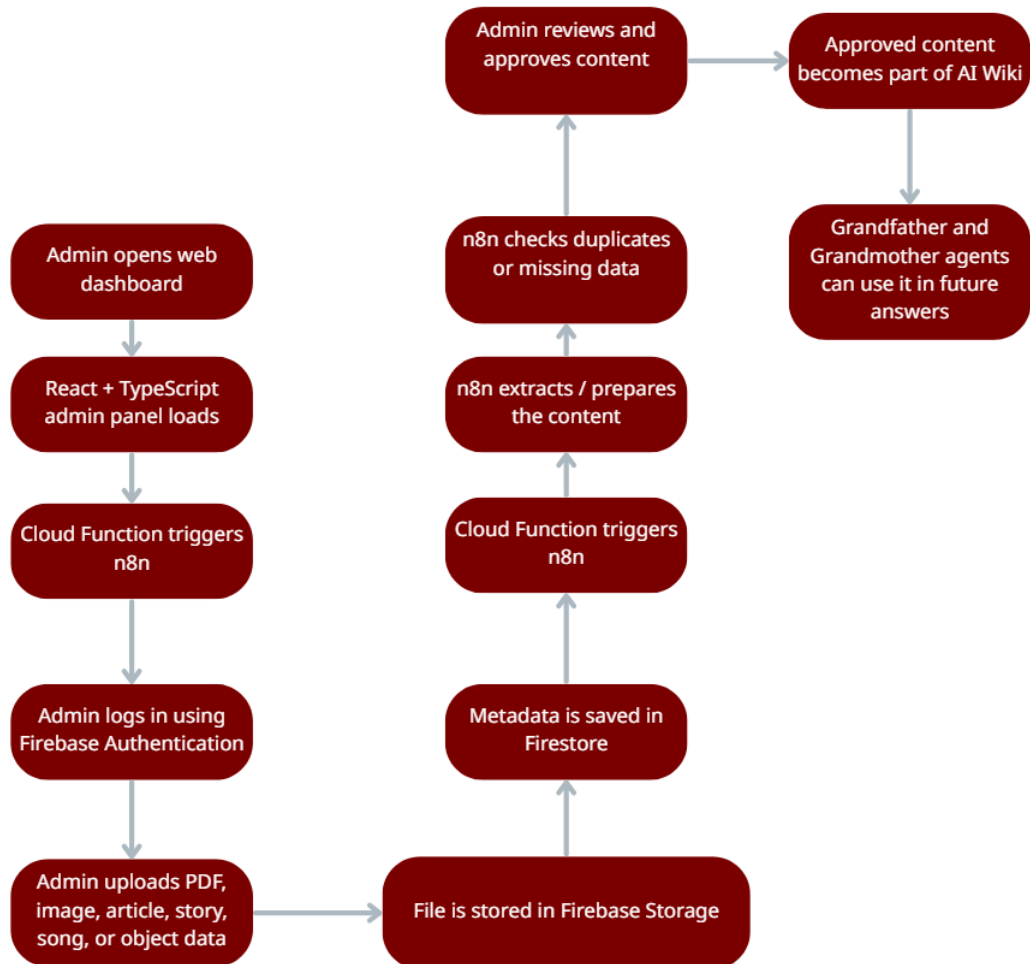


Figure 3.1 4

System Flow Samary

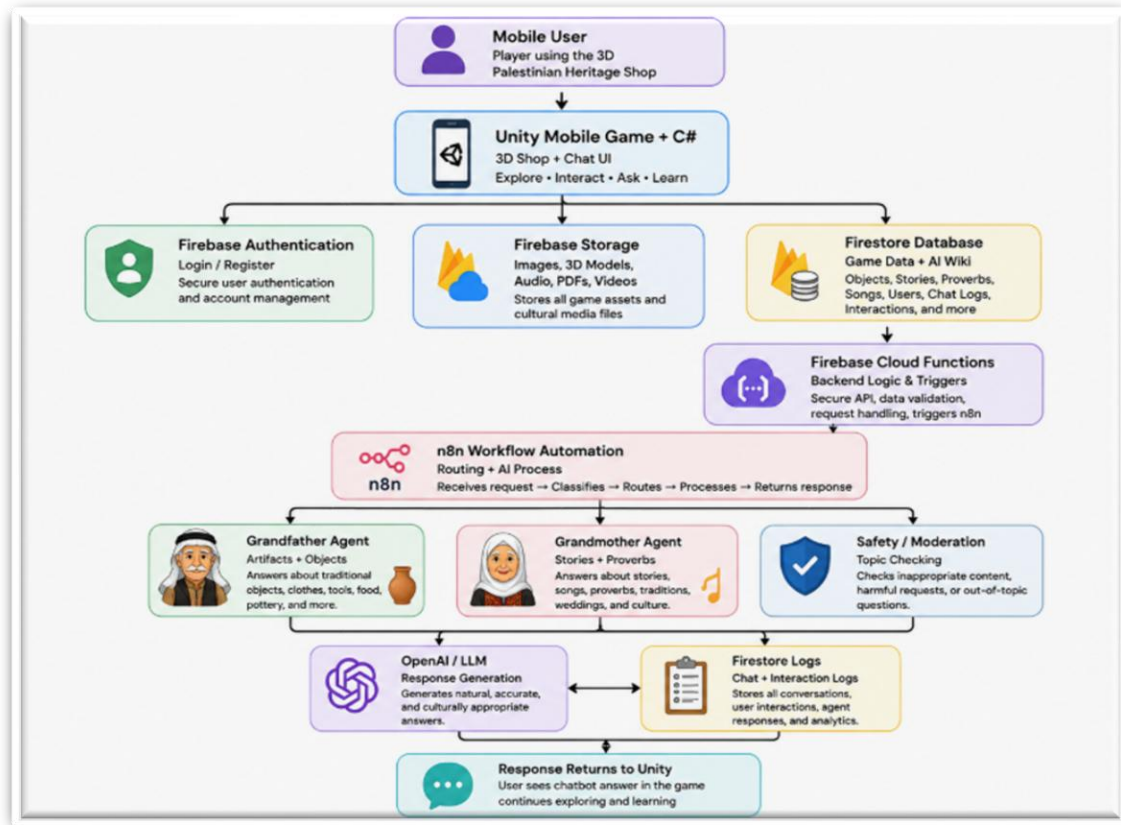


Figure 3.1 5

(generated by tool nano banana)

System Flow Samary(human made)

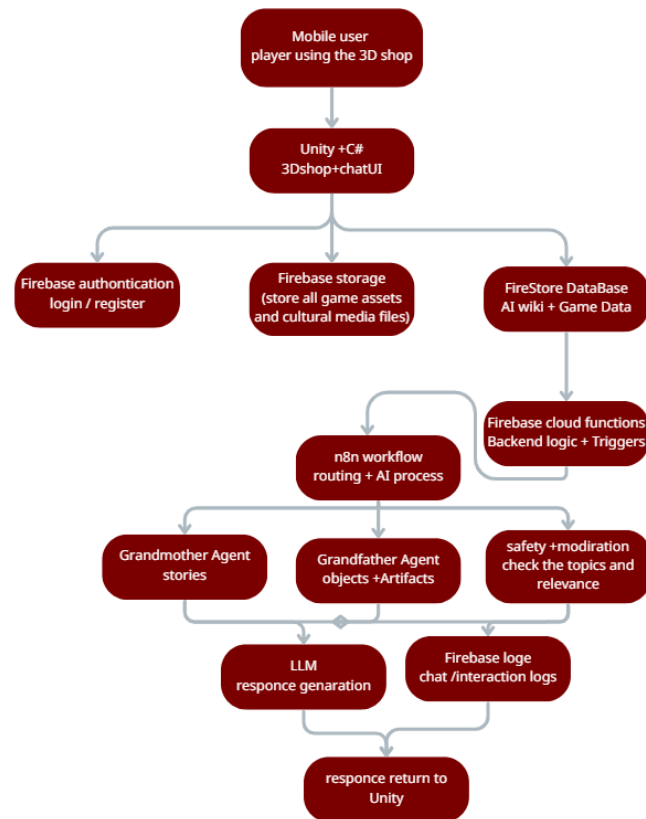


Figure 3.16

3.3 Post-Production (Launch & Support)

Launch:

Releasing the game on the target platform (Mobile). Coss platform goal is also in mind for future work.

Maintenance:

Developing patches, updates, and additional content (DLC) based on user's feedback as future work.

3.4 Story

Complete User Interaction Inside the Palestinian Heritage Virtual Shop

Actors

1. User / Visitor
2. Grandfather Chatbot
3. Grandmother Chatbot
4. AI Wiki System
5. 3D Environment System
6. Admin

A user opens the application and enters a virtual 3D Palestinian heritage shop designed to simulate an old Palestinian shop.

The environment contains traditional Palestinian clothes, handmade tools, decorations, pottery and historical cultural artifacts representing Palestinian heritage.

When the app starts, the system loads the virtual shop environment and activates the chatbots icon or avatar.

The user starts exploring the app using controls.

The user can:

1. explore the environment
2. Interact with cultural objects
3. Open information windows
4. Ask questions through chatbots

As a user see a traditional Palestinian embroidered thobe then can clicks on the thobe to learn more about it.

The interaction is detected by 3D environment system which can sends the information about the object to the grandfather chatbot.

The grandfather chatbot analyses the selected object and requests related information from the AI wiki system.

The AI wiki search in its cultural database and retrieves detailed information including:

1. Historical background
2. Region of origin
3. Cultural symbolism
4. Traditional uses
5. Materials used in production
6. Importance in Palestinian identity

The grandfather chatbot process the information and responds the personality and speaking style of an elderly Palestinian shop owner who has deep knowledge about Palestinian cultural artifacts and traditions.

The chatbot response appears in a conversational screen in a conversational format.

The user asks:

“موجود قصص متعلقة بالعرس الفلسطيني يا جدي؟”

The grandfather chatbot analyses the question and decides that request is related to the stories and traditions rather than physical artifacts. The request is then transferred to the grandmother chatbot.

The Grandmother Chatbot retrieves related information from the

AI Wiki System including:

1. Wedding traditions
2. Palestinian folk stories
3. Traditional songs
4. Cultural sayings
5. Celebration customs

The grandmother chatbot respond in a worm storytelling style using traditional Palestinian sayings.

The user continues the conversation and asks like this:

اخبرني اكثر عن اللبس في الاعراس
الاهازيج او الأغاني الشعبية في الاعراس
في أي مدن تم استخدام هذه التقاليد

The system identifies the object as an oud instrument and sends the request to the Grandfather Chatbot.

The chatbot retrieves information from the AI wiki about:

1. The history of the oud
2. Traditional Palestinian music
3. Cultural importance of instruments

The user asks:

“ما هو اكثر شئ مهم في التراث الفلسطيني؟”

Both chatbots generate possible responses based on their knowledge and speaking style. The system evaluates the response and select the most relevant answer before displaying to the user

The user can continue exploring the environment and interact with:

1. Traditional clothing
2. Historical decorations
3. Folk stories
4. Proverbs and sayings

At one point, the user asks a question unrelated to the Palestinian heritage.

The system analyzes the request and politely sends to the user

“المعذرة هذا النظام متخصص في التراث الفلسطيني”

At the same time, the admin accesses the management panel of the AI wiki system.

The admin performs maintenance operations such as:

1. Adding new cultural articles
2. Updating stories and songs
3. Editing object descriptions
4. Uploading historical references
5. Correcting inaccurate information
6. Expanding the knowledge database

The updated information immediately becomes available to both chatbots.

throughout the entire interaction, the system continuously performs:

1. User input analysis
2. Cultural information retrieval
3. Chatbot communication
4. Context tracking
5. Dynamic response generation
6. Object interaction processing

The application maintains:

1. Accurate cultural information
2. Fast response time
3. Smooth navigation
4. Consistent chatbot personalities
5. Interactive user experience
6. Immersive cultural environment

At the end, the user exits the finishing the journey that combines the Palestinian heritage, storytelling, AI, interactive cultural exploration inside virtual 3D environment

3.5 Scenarios

Scenario 1: Opening the App and Loading the Virtual Store

1. **User/Visitor**: Opens the app and enters a 3D Palestinian Heritage store.
2. **Palestinian Heritage App**: Starts the user session and prepares the login interface.
3. **3D Environment System**: Loads the store's traditional environment and displays clothing, tools, pottery, decorations, and historical artifacts.

Scenario 2: Activating Characters and Starting Exploration

1. **3D Environment System**: Displays chatbot icons or avatars within the store.
2. **User/Visitor**: Moves within the virtual environment using controls and explores items.
3. **Information Popup**: Becomes available when a cultural item is selected

Scenario 3: Selecting a Palestinian Dress

1. **User/Visitor**: Sees an embroidered Palestinian dress and clicks on it to learn its details.
2. **Cultural Artifact**: The dress is represented as an interactive cultural object within the store.
3. **3D Environment System**: Detects the click event on the dress and identifies the selected item's data.

Scenario 4: Sending Item Data to the Store Grandfather

1. **3D Environment System:** Sends information about the selected item to the Grandfather Chatbot.
2. **Grandfather Chatbot:** Analyses the selected item and requests related cultural information.
3. **AI Wiki System:** Receives a request to search for information about the Palestinian dress.

Scenario 5: Retrieving Dress Information from AI Wiki

1. **AI Wiki System:** Searches the cultural database and retrieves the
 - a. historical background, region of origin, cultural symbolism, traditional
 - b. uses, materials, and significance of the item in Palestinian identity.
2. **Cultural Information:** The retrieved information contains structured and unstructured data about the item.
3. **Grandfather Chatbot:** Processes the information before displaying it to the user.

Scenario 6: Grandfather's Response in the Style of an Old Palestinian Shopkeeper

1. **Grandfather Chatbot:** Responds as an elderly Palestinian man with deep knowledge of antiques and traditions.
2. **Information Popup:** Displays the response in an understandable chat screen or information window.
3. **User/Visitor:** Reads the response and learns about the Palestinian thobe.

Scenario 7: User Asks About Palestinian Wedding Stories

1. **User/Visitor:** Asks, “Grandpa, are there any stories related to Palestinian weddings?”
2. **Grandfather Chatbot:** Analyses the question and realizes it pertains to stories and traditions, not physical artifacts.

Scenario 8: Grandmother's Warm, Storytelling Response

1. **Grandmother Chatbot:** Requests information about Palestinian weddings from the AI Wiki.
2. **AI Wiki System:** Retrieves wedding traditions, folk tales, songs, proverbs, and celebration customs.
3. **Grandmother Chatbot:** Responds in a warm, storytelling style reminiscent of grandmothers' tales, using Palestinian folk expressions.

Scenario 9: Follow-up Questions about Weddings

1. **User/Visitor:** Asks about wedding attire, folk songs or chants, and the cities that practice these traditions.
2. **Grandmother Chatbot:** Maintains the conversation flow and retrieves relevant additional details.
3. **AI Wiki System:** Provides the chatbot with up-to-date information about customs and regions.

Scenario 10: Interacting with the Oud Instrument

1. **User/Visitor:** Interacts with the oud instrument inside the store.
2. **3D Environment System:** Identifies the item as an oud instrument and sends a request to the Grandfather Chatbot.
3. **Grandfather Chatbot:** Retrieves the history of the oud, Palestinian music, and the importance of musical instruments from the AI Wiki.
4. **Router/Agent Assignment:** Directs the request to the Grandmother Chatbot, as it specializes in stories, songs, and proverbs.

Scenario 11: A general question about the most important thing in Palestinian heritage

1. **User/Visitor:** Asks: "What is the most important thing in Palestinian heritage?"
2. **Grandfather Chatbot:** Generates a possible answer from the perspective of artifacts and material identity.
3. **Grandmother Chatbot:** Generates a possible answer from the perspective of stories, customs, and popular memory.
4. **Response Evaluation System:** Compares the answers and selects the most appropriate one before displaying it to the user.

Scenario 12: Continued Exploration and Recommendations

1. **User/Visitor:** Continues exploring clothing, decorations, stories, proverbs, and songs.
2. **Interaction History:** Records items and questions the user has been interested in.
3. **Recommendation System:** Suggests items, stories, or songs related to the user's preferences.

Scenario 13: Question Outside the Scope of Palestinian Heritage

1. **User/Visitor:** Submits a question unrelated to Palestinian heritage.
2. **Content Moderation System:** Analyses the question and confirms it is outside the scope of the system or inappropriate.
3. **Palestinian Heritage App:** Displays a polite message: "Sorry, this system specializes in Palestinian heritage."

Scenario 14: AI Wiki Content Management

1. **Admin:** Accesses the AI Wiki admin panel.
2. **AI Wiki System:** Allows adding articles, updating stories and songs, editing item descriptions, uploading historical references, and correcting inaccurate information.
3. **Admin:** Reviews the content and ensures its cultural accuracy before approval.

Scenario 15: Live Chatbot Information Update

1. **AI Wiki System:** Makes updated information instantly available to Grandparents.
2. **Grandfather Chatbot:** Uses the new version when answering questions about antiques and tools.
3. **Grandmother Chatbot:** Uses the new version when answering questions about stories, songs, and traditions.

Scenario 16: Ongoing System Operations During the Journey

1. **Palestinian Heritage App**: Manages context tracking, response generation, and interaction processing.
2. **3D Environment System**: Handles object and movement events within the store.
3. **Router/Agent Assignment**: Selects the appropriate agent based on the type of request.

Scenario 17: Ending the Journey

1. **User/Visitor**: Ends the experience after a journey that combines Palestinian heritage, stories, artificial intelligence, and 3D interactive exploration.
2. **Palestinian Heritage App**: Closes the session and saves the interaction for future reference.
3. **Interaction History**: Maintains a record suitable for future recommendations.

Scenario 18: Unregistered User Experience

- 1.
2. **User/Visitor**: a guest can interact with the items in the shop like the previous scenarios but in limited amount of times
3. **Grandmother/Grandfather chatbots**: can reply to any question the guest has but the user has limit about of massages
4. **AI Wiki System**: will provide any requested information when the agents request for ones

3.6 System Requirements

Functional requirements

User interaction requirements

1. It should enable the user to browse through the virtual 3D Palestinian heritage shop.
2. It should enable the user to navigate through the virtual environment by moving around within the environment.
3. It should enable the user to interact with the cultural artifacts available in the shop.
4. It should enable the user to select any cultural artifact for more information by clicking on it.
5. It should display information pop-ups of the selected cultural artifact.

Requirements for the Chatbot

1. The system should have a Grandfather Chatbot that specializes in the artifacts and objects of Palestinian culture.
2. The system should have a Grandmother Chatbot that specializes in Palestinian tales, traditions, songs, and sayings.
3. The system should let the user talk with the chatbots via an interactive interface.
4. The system should analyze the user's question and find out which chatbot to reply to.
5. The system should transfer the user's interaction between the chatbots when the question is about something else.
6. The system should provide chatbot replies that match their personalities.

AI Wiki Requirements

1. The system shall retrieve cultural information from the AI Wiki database.
2. The AI Wiki shall store structured and unstructured Palestinian cultural information.
3. The system shall retrieve:
 - a. Historical background
 - b. Traditional stories
 - c. Object descriptions
 - d. Cultural symbolism
 - e. Regional information
4. The system shall update chatbot responses using the latest AI Wiki information.

Response Evaluation Requirements

1. The system should assess various responses from the chatbot if required.
2. The system should select the most relevant response before presenting it to the user.
3. The system should not produce irrelevant responses concerning Palestinian culture.
4. The system should alert the user if the query falls outside the domain of the system.

Recommendation Requirements

1. The system must record the users' interaction history within the application.
2. The system will recommend culturally related items based on the users' preferences.
3. The system will recommend related stories, songs, and artifacts while exploring.

Administration Requirements

1. The software must feature an admin management console.
2. The admin must have the capability of adding cultural data into the AI Wiki.
3. The admin must have the capability of editing descriptions and historical facts.
4. The admin must have the capability of updating stories, songs, and customs.
5. The admin must have the capability of correcting inaccuracies present in the database.
6. The software must make the updated data immediately accessible by both chatbots.

Requirements for System Processing

1. The system should handle user input continuously while interacting.
2. The system should enable real-time communication with chatbots.
3. The system should handle events of interaction with objects in the 3D environment.
4. The system should dynamically generate responses from chatbots.
5. The system should ensure personality consistency of chatbots throughout the discussion.

Non-Functional Requirements

Performance Requirements

1. The application shall guarantee quick responses during chats with the chatbot.

2. The system shall be able to load the 3D environment promptly without delays.
3. The system shall allow smooth movement in the virtual environment.
4. The system shall enable real-time interaction processing.

Usability Requirements

1. The system shall offer an easily usable interface by people of various ages.
2. The interface of the chatbot shall be easy to use.
3. The system shall offer an engaging experience for cultural interactions.
4. The conversational responses generated by the system shall remain clear and readable.

Accuracy Requirements

1. The system must deliver accurate information about the Palestinian culture.
2. The system must not generate false or irrelevant answers.
3. The chatbot avatars must maintain consistency in all interactions.

Scalability Requirements

1. The system must enable the addition of new cultural information and artifacts in the future.
2. The AI Wiki must enable continued growth of the knowledge base.
3. The system architecture must fit the addition of new chatbot characters in subsequent releases.

Security Requirements

1. The system will limit access to the admin management interface.
2. The system will protect the AI Wiki database against any unauthorized alterations.
3. The system will ensure secure storage of cultural information and system information.

Maintainability Requirements

1. The system will facilitate the simple update of stories, objects, and cultural information.
2. The system will provide for the maintenance and adjustment of the chatbot's conversation patterns.
3. The system will enable effective management of cultural content by the administrators.

Compatibility Requirements

1. The application should be compatible with supported desktop operating systems.
2. The system should support the capability of integrating with 3D virtual environment technology.

3. The system should support the interaction and display of content using Arabic.

Cultural Experience Requirements

1. The system should continue the Palestinian cultural identity through its interface design and conversation.
2. The system should offer an immersive storytelling experience.
3. The system should sustain authentic Palestinian expressions and culture in the chatbot conversations.
4. The system should provide an enriching education experience on the Palestinian culture.

3.7 UML Design

Context Diagram

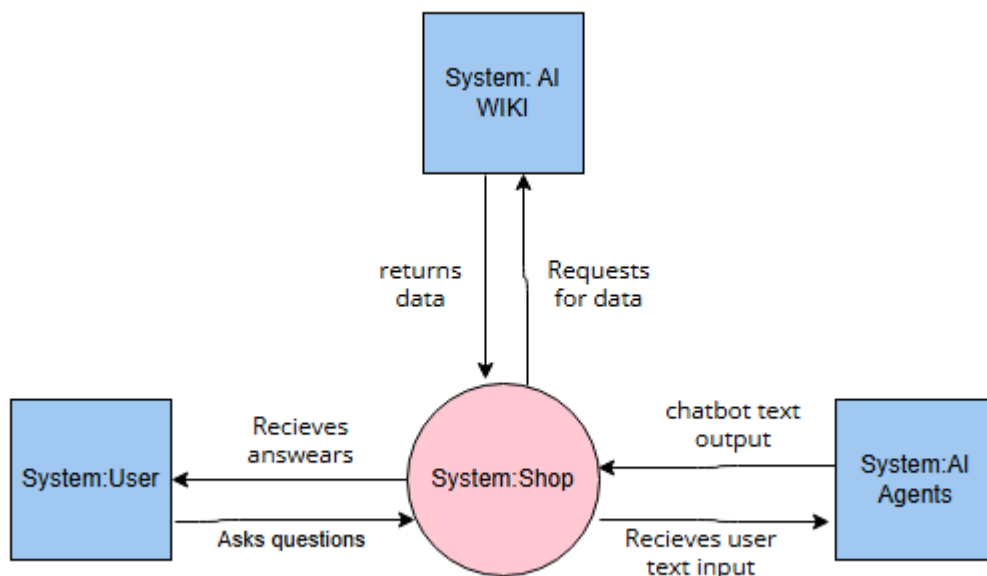


Figure 3.1 7

The context diagram illustrates the overall interaction between the **Shop System** and its external entities.

Use Case Diagrams

Use case diagram for External users

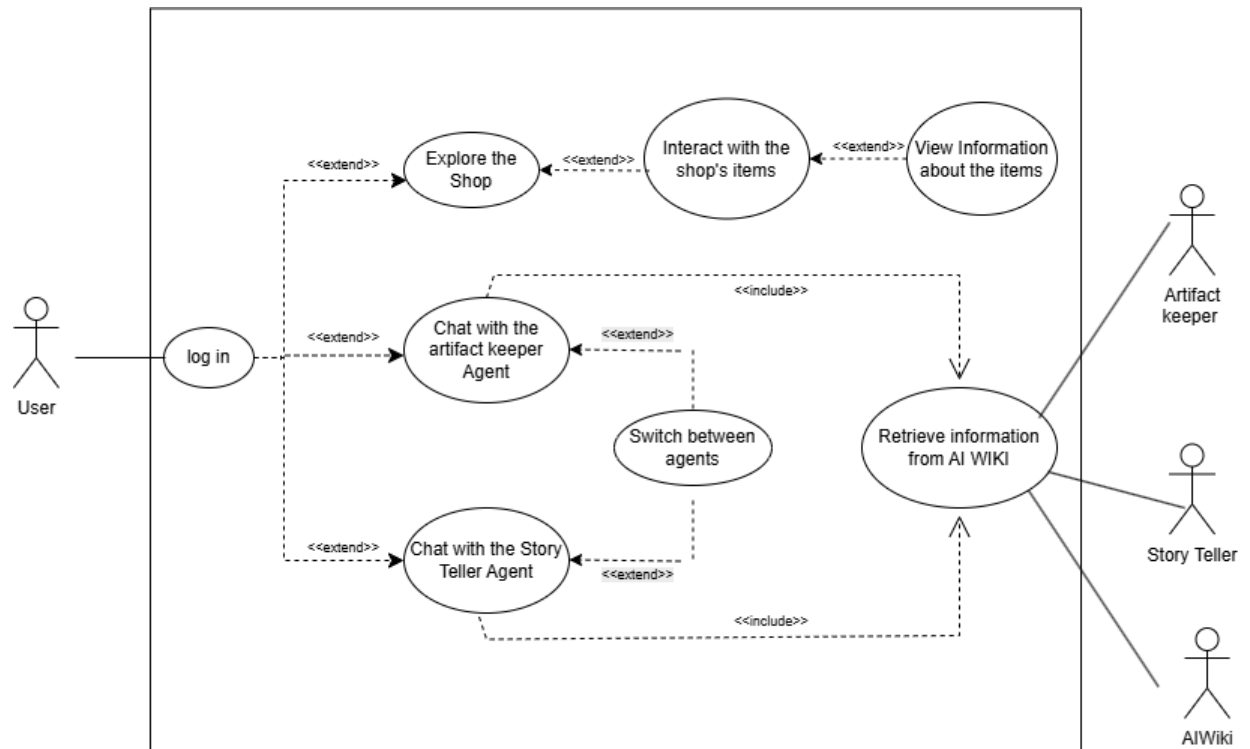


Figure 3.1 8

External users Use Case Relationships

Source Use Case	Target Use Case	Relationship Type
Log in	Explore the Shop	<<extend>>
Log in	Chat with the Artifact Keeper Agent	<<extend>>
Log in	Chat with the Story Teller Agent	<<extend>>
Interact with the Shop's Items	Explore the Shop	<<extend>>
View Information about the Items	Interact with the Shop's Items	<<extend>>
Switch between Agents	Chat with the Artifact Keeper Agent	<<extend>>
Switch between Agents	Chat with the Story Teller Agent	<<extend>>
Chat with the Artifact Keeper Agent	Retrieve Information from AIWIKI	<<include>>
Chat with the Story Teller Agent	Retrieve Information from AIWIKI	<<include>>

Table 3.1 2

Use case diagram for AI WIKI

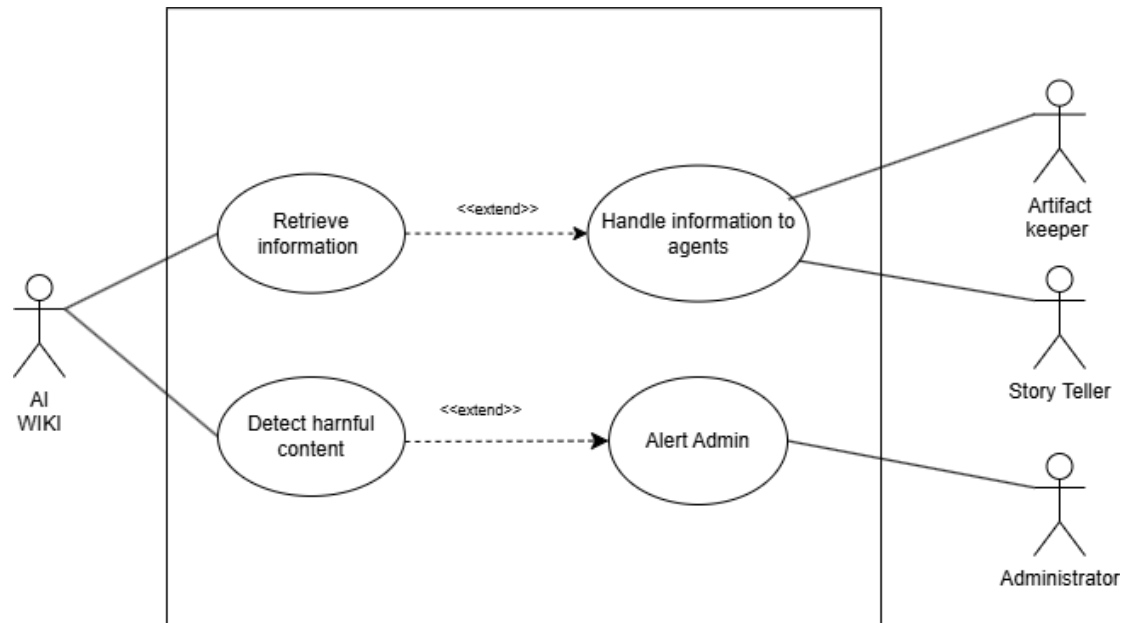


Figure 3.1 9

AIWiki Use Case

Source Use Case	Target Use Case	Relationship
AIWiki	Retrieve Information	<<extend>>
AIWiki	Detect harmful content	<<extend>>
Retrieve Information	Handle information to agents	<<include>>
Detect harmful content	Alert admin	<<include>>

Table 3.1 3

Use case diagram for AI Agents

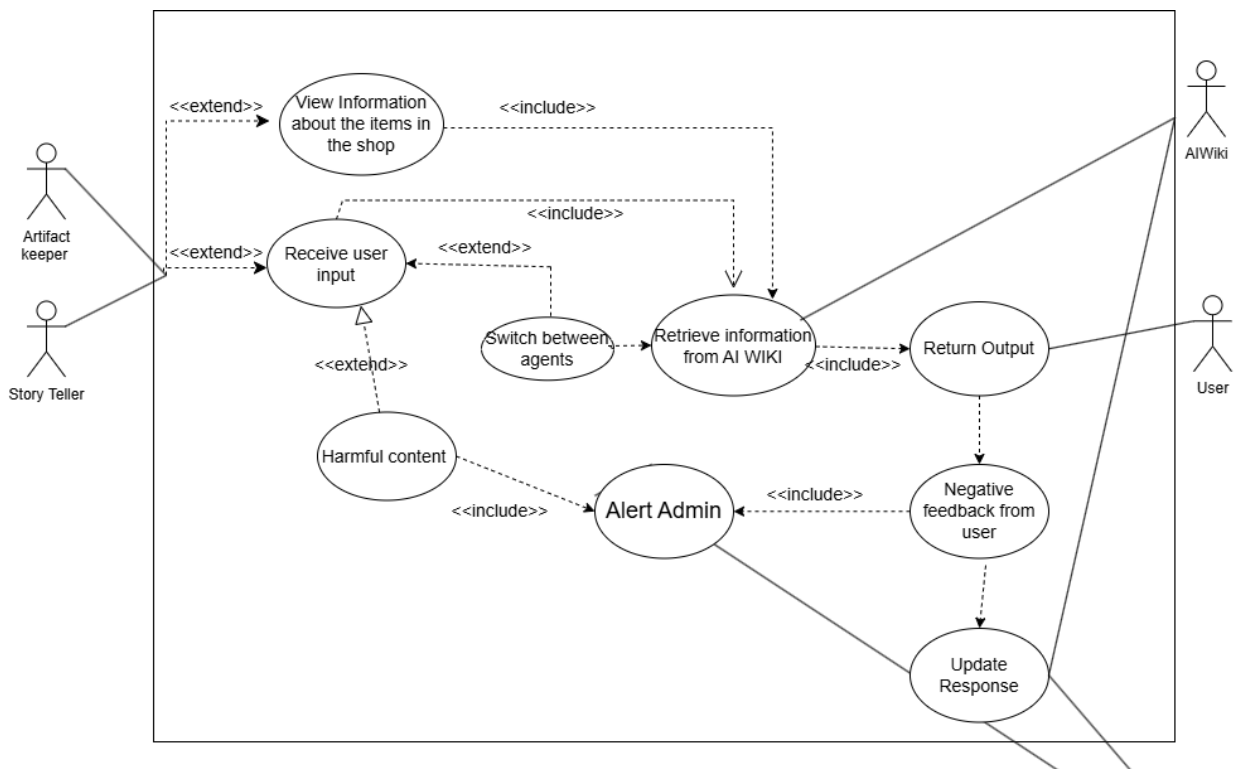
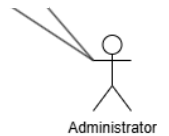


Figure 3.1 10



AI Agents Use Case

Source Use Case	Target Use Case	Relationship
View information about the items in the shop	Retrieve information from AI WIKI	<<include>>
Receive user input	Retrieve information from AI WIKI	<<include>>
Switch between agents	Receive user input	<<extend>>
Retrieve information from AI WIKI	Return Output	<<include>>
Return Output	Negative feedback from user	Association
Negative feedback from user	Alert Admin	<<include>>
Harmful content	Alert Admin	<<include>>
Negative feedback from user	Update Response	Association

Table 3.1 4

Use case diagram for AI Administrator

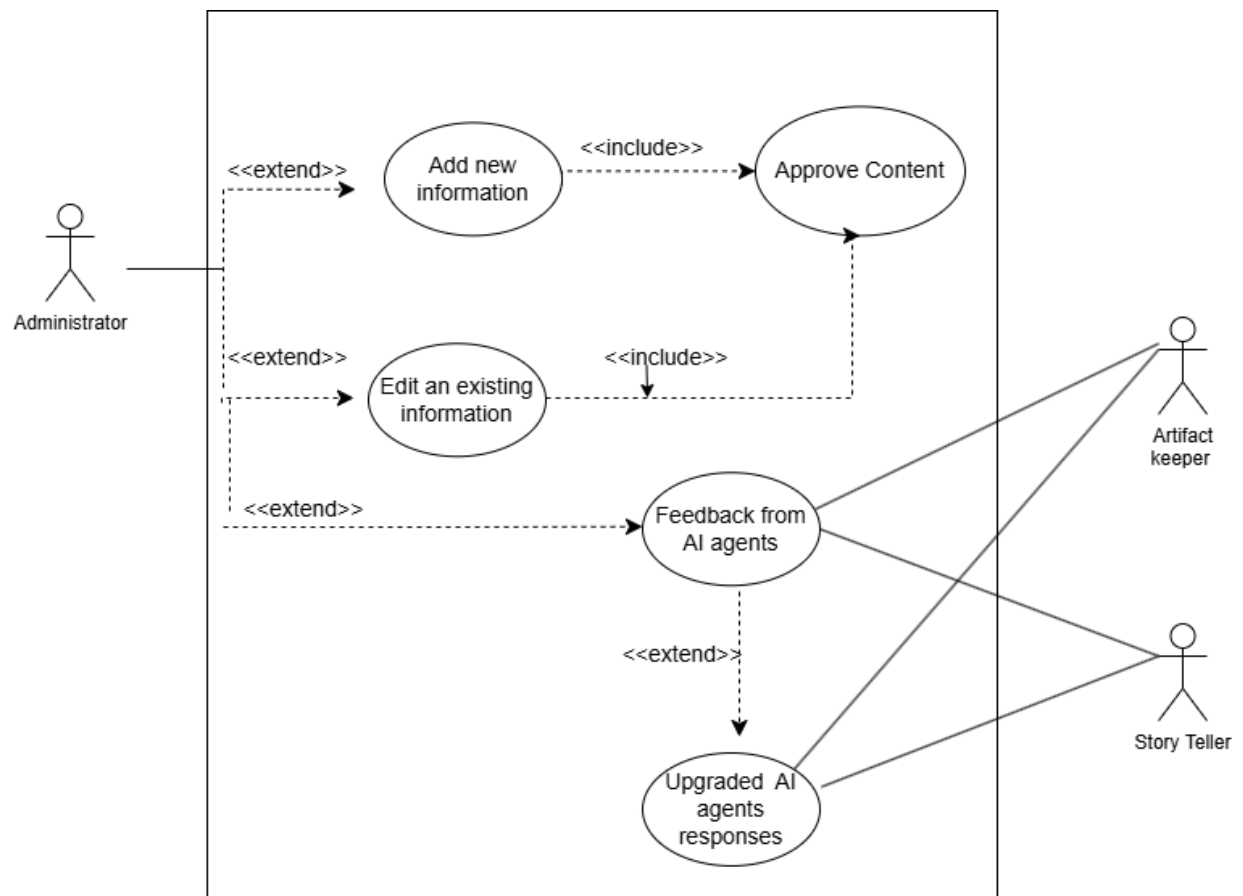


Figure 3.1 11

Administrator Use Case

Source Use Case	Target Use Case	Relationship
Administrator	Add new information	<<extend>>
Administrator	Feedback from AI agents	<<extend>>
Administrator	Edit an existing information	<<extend>>
Add new information	Approve Content	<<include>>
Edit an existing information	Approve Content	<<include>>
Feedback from AI agents	Upgrade AI agents responses	<<extend>>

Table 3.1 5

Sequence Diagrams

Registration process

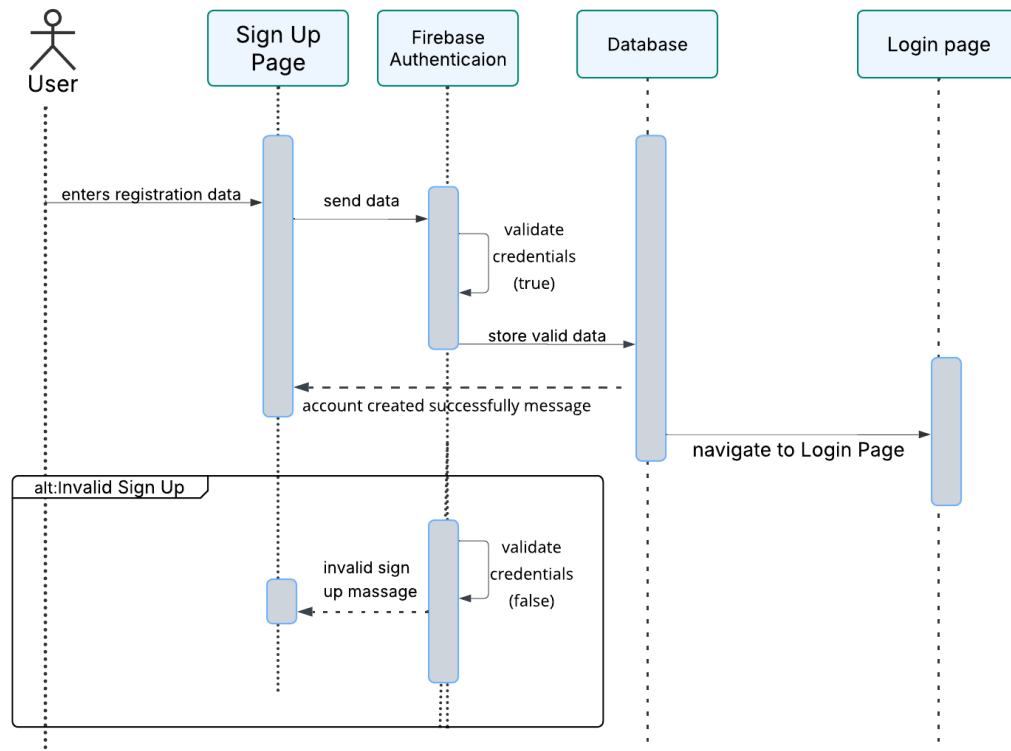


Figure 3.1 12

1. Registration data are:
 - Username
 - email
 - password
 - confirm password
2. Validate credentials details:
 - required fields
 - email format
 - passwords match
 - email not already exists

Login process

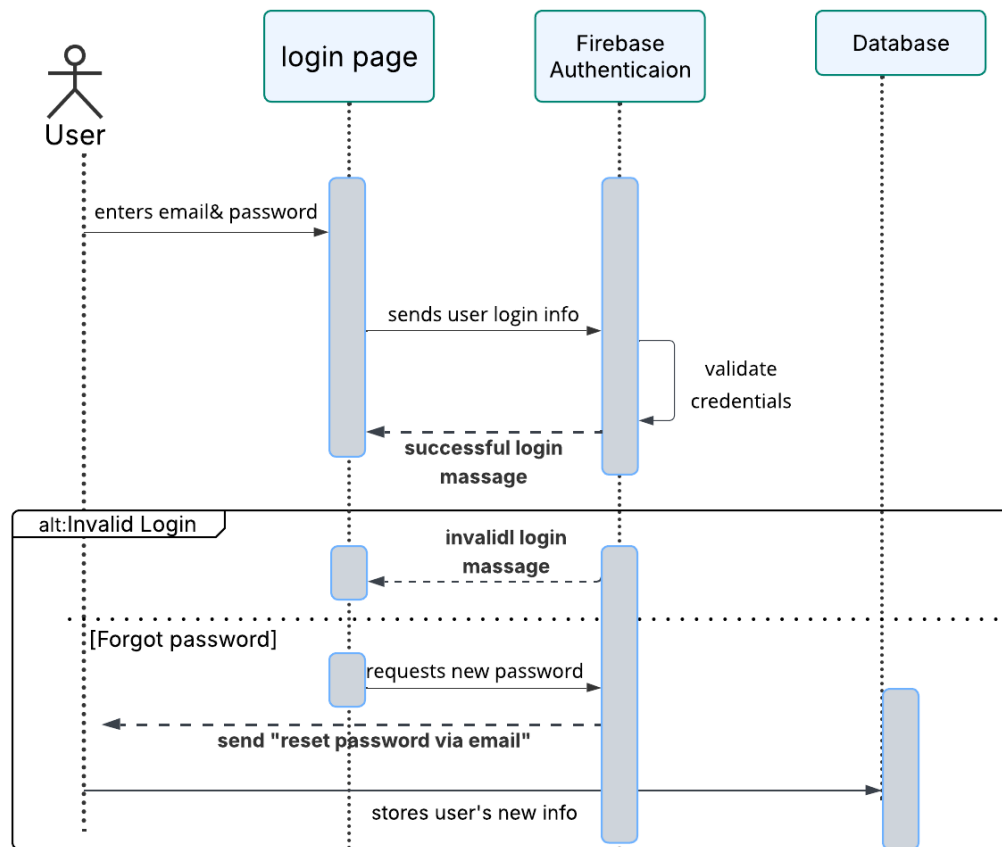


Figure 3.1 13

Registered User's Interactions with the system

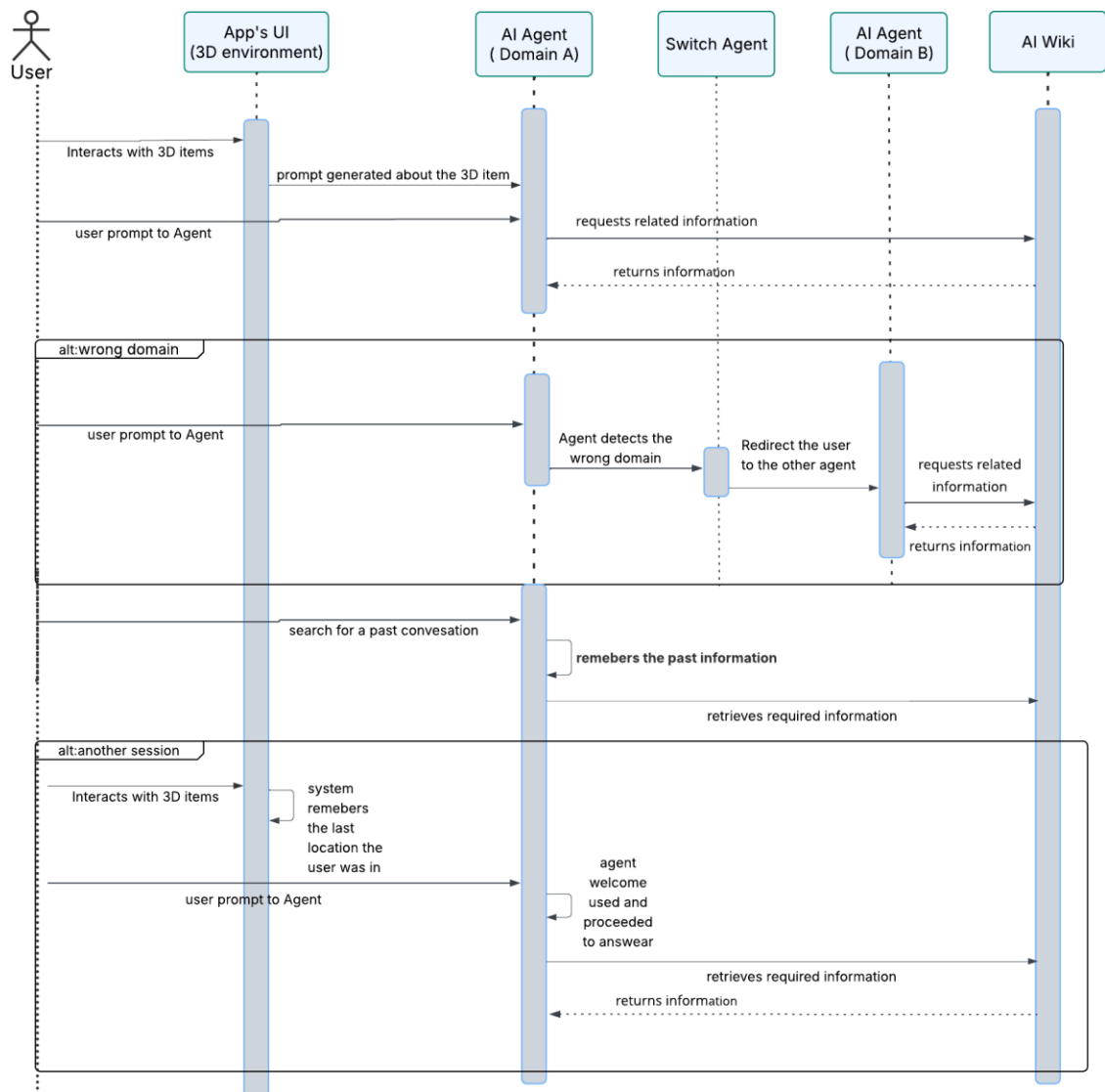


Figure 3.1 14

Unrejected user

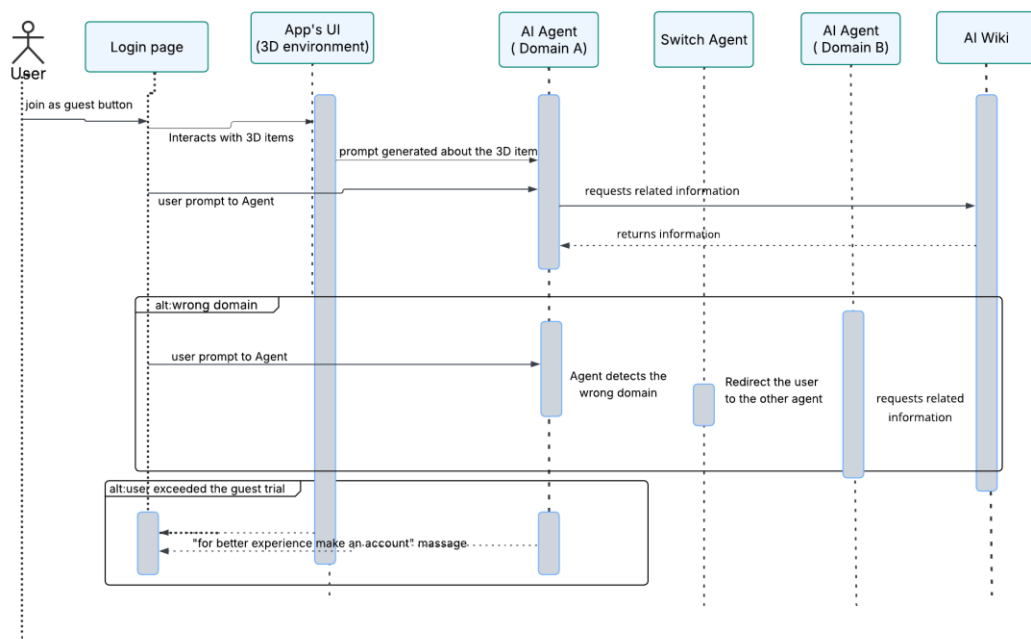


Figure 3.1 15

Wiki Admin

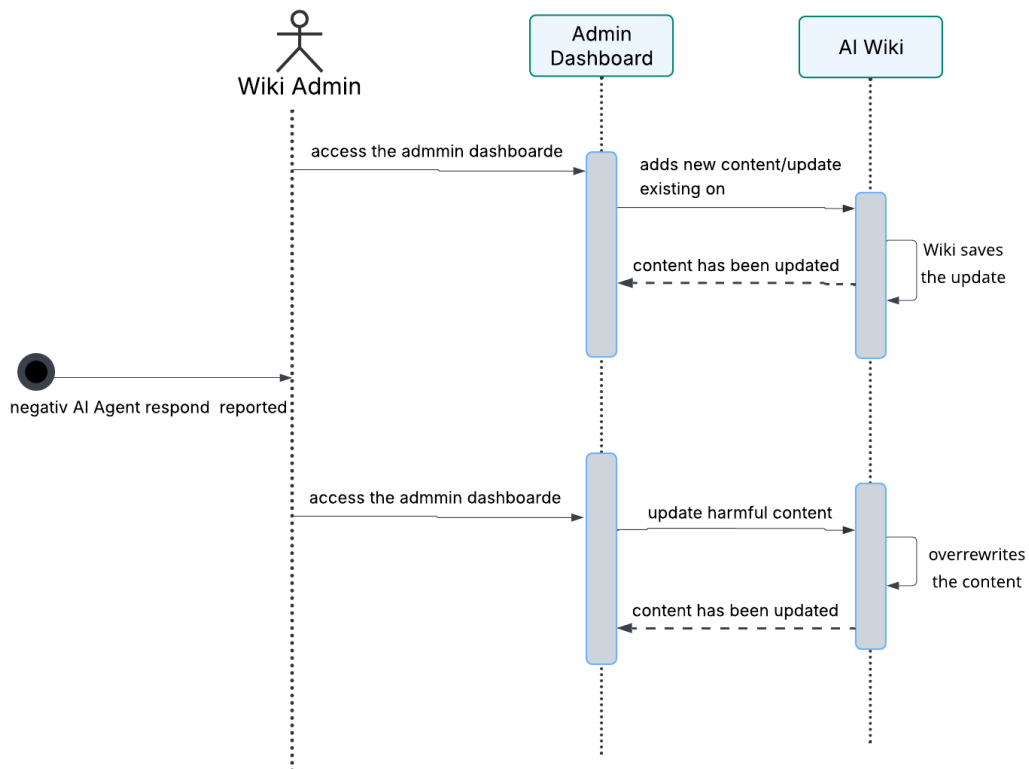


Figure 3.1 16

The UML diagrams presented in this chapter were created using Visual Paradigm Online, draw.io, and Lucidchart [16]–[18].

Chapter 4: System Design

4.1 Class Diagram

This diagram will help to explain how the system elements interact with one another from a traditional programming perspective

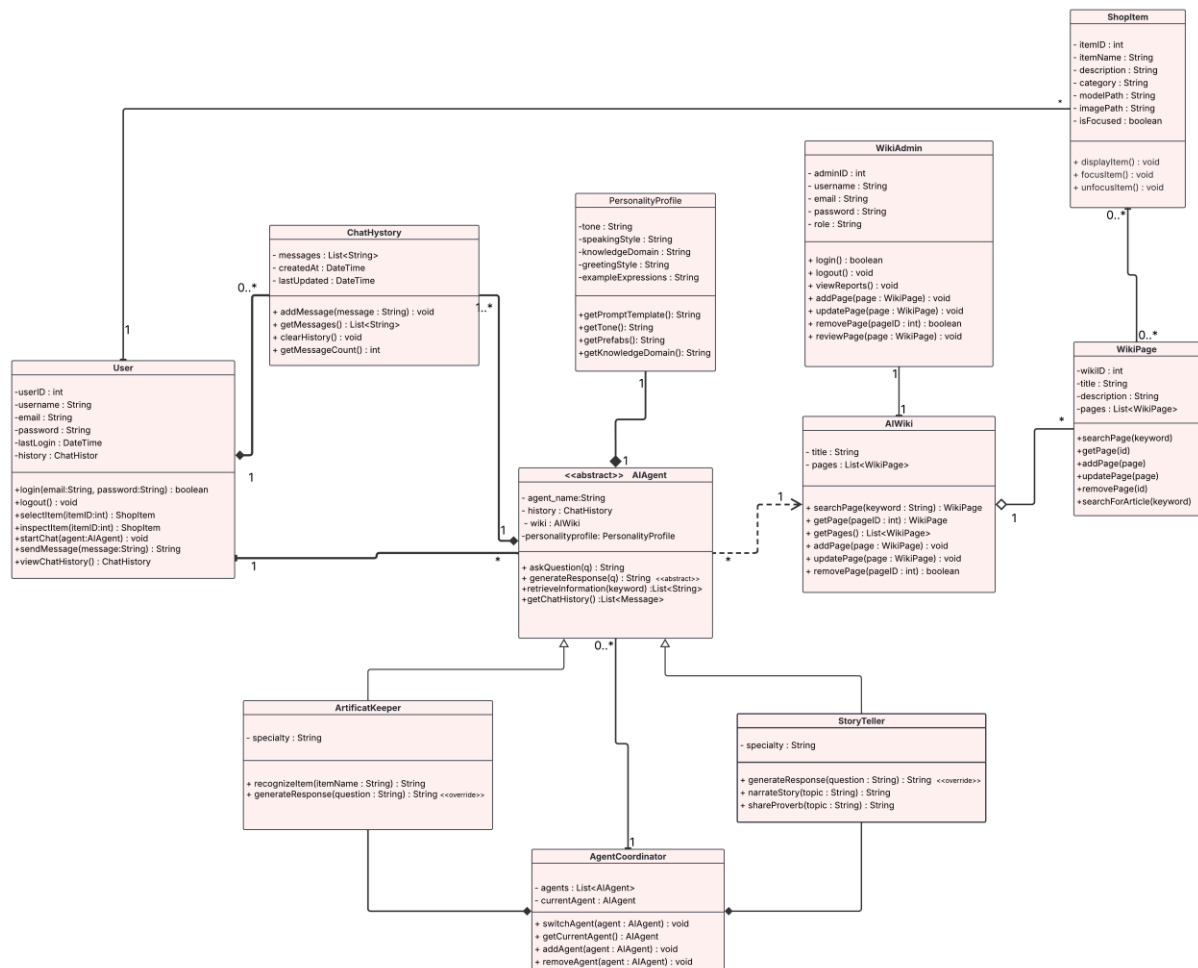


Figure 4.1 1

UML Class Summary

Table 1: Classes and Descriptions

Class Name	Description
AIAgent (Abstract)	Base AI agent class that provides shared AI functionality and defines specialized agent behavior.
Storyteller	Specialized AI agent for storytelling, proverbs, and narrative responses.
ArtifactKeeper	Specialized AI agent that recognizes and provides information about shop items and artifacts.
AgentCoordinator	A central controller that manages multiple AI agents and switches between them
PersonalityProfile	Stores personality settings such as tone, speaking style, greeting style, and knowledge domain.
ChatHistory	Stores conversation records and previous interactions.
AIWiki	Knowledge source used by AI agents to retrieve and organize information.
WikiAdmin	Administrator responsible for maintaining and supervising wiki content.
WikiPage	Represents a wiki page containing information related to items and topics.
ShopItem	Represents an item available in the shop.
User	End user interacting with AI agents and shop items.

Table 4.1 1

Table 2: Class Relationships

Class A	Class B	Relationship	Multiplicity	Explanation
AIAgent	PersonalityProfile	Composition	1 ↔ 1	Each AI agent owns exactly one personality profile, and each personality profile belongs to one AI agent.
AIAgent	Storyteller	Inheritance / Generalization	No multiplicity	StoryTeller is a specialized type of AIAgent.
AIAgent	Artifact Keeper	Inheritance / Generalization	No multiplicity	ArtifactKeeper is a specialized type of AIAgent.
AIAgent	AIWiki	Association	* ↔ 1	Multiple AI agents can access the same AIWiki knowledge source.
AIAgent	ChatHistory	Composition	1 ↔ 1	Each AI agent maintains one chat history containing its conversation records.
User	AIAgent	Association	1 ↔ *	One user can talk to multiple agents.
User	ChatHistory	Composition	1 ↔ 0..*	A user may have multiple chat histories over time.
User	ShopItem	Association	* ↔ *	A user can browse, inspect, and interact with many shop items
AIWiki	WikiAdmin	Association	1 ↔ 1	One AIWiki is managed by one Administrator.
AIWiki	WikiPage	Dependency	1 ↔ *	An AIWiki contains multiple wiki pages.
WikiPage	ShopItem	Association	* ↔ *	A wiki page may describe multiple items, and an item may appear on multiple pages.
AgentCoordinator	AIAgent	Association	1 ↔ *	The coordinator manages multiple AI agents and can switch between them.

Table 4.1 2

The presented ERD shows a cultural chatbot platform in which users register an account and begin chats. Sessions can include more than one conversation with chatbots. Conversations consist of messages from either the users or the chatbot itself. Chatbot responses are rated via the ResponsesEvaluations table. In addition, there is an ability to engage in interaction with different cultural objects, such as artifacts, items, pictures, or even 3D models. All interactions are logged in the Interactions table; based on this table, the platform generates recommendations regarding the chosen cultural artifacts or related cultural information. These artifacts are linked with cultural information via the ArtifactInformation table, meaning one artifact has multiple cultural descriptions. Admins are involved in cultural information and system content management; all their activities are stored in the AdminActivityLogs table. Messages considered inappropriate are registered in the ModerationLogs table as shown:



Chapter 5: Testing

5.1 Verification and Validation Requirements

Functional requirements validation	
User interaction Requirements	
Requirement	Done(Y/N)
It should enable the user to browse through the virtual 3D Palestinian heritage shop.	Y
It should enable the user to navigate through the virtual environment by moving around within the environment.	Y
It should enable the user to interact with the cultural artifacts available in the shop.	Y
It should enable the user to select any cultural artifact for more information by clicking on it.	Y
It should display information pop-ups of the selected cultural artifact.	Y
Requirements for the Chatbot	
The system should have a Grandfather Chatbot that specializes in the artifacts and objects of Palestinian culture.	Y
The system should have a Grandmother Chatbot that specializes in Palestinian tales, traditions, songs, and sayings.	Y
The system should let the user talk with the chatbots via an interactive interface.	Y

The system should analyze the user's question and find out which chatbot to reply to.	Y
The system should transfer the user's interaction between the chatbots when the question is about something else.	Y
The system should provide chatbot replies that match their personalities.	Y
AI Wiki Requirements	
The system shall retrieve cultural information from the AI Wiki database.	Y
The AI Wiki shall store structured and unstructured Palestinian cultural information.	Y
The system shall retrieve: a. Historical background b. Traditional stories c. Object descriptions d. Cultural symbolism e. Regional information	Y
The system shall update chatbot responses using the latest AI Wiki information.	Y
Response Evaluation Requirements	
The system should assess various responses from the chatbot if required.	Y
The system should select the most relevant response before presenting it to the user.	Y
The system should not produce irrelevant responses concerning Palestinian culture.	Y
	Y

The system should alert the user if the query falls outside the domain of the system.	
Recommendation Requirement	
The system must record the users' interaction history within the application.	Y
The system will recommend culturally related items based on the users' preferences.	Y
The system will recommend related stories, songs, and artifacts while exploring.	Y
Administration Requirements	
The software must feature an admin management console	Y
The admin must have the capability of adding cultural data into the AI Wiki.	Y
The admin must have the capability of editing descriptions and historical facts.	Y
The admin must have the capability of correcting inaccuracies present in the database.	Y
The software must make the updated data immediately accessible by both chatbots.	Y
Requirements for System Processing	
The system should handle user input continuously while interacting.	Y
The system should enable real-time communication with chatbots.	Y
The system should handle events of interaction with objects in the 3D environment.	Y

The system should dynamically generate responses from chatbots.	Y
The system should ensure personality consistency of chatbots throughout the discussion.	Y

Table 5.1 1

Non-Functional requirements validation	
Requirements	Done(Y/N)
Performance Requirements	
The application shall guarantee quick responses during chats with the chatbot.	Y
The system shall be able to load the 3D environment promptly without delays.	Y
The system shall allow smooth movement in the virtual environment.	Y
The system shall enable real-time interaction processing.	Y
Usability Requirements	
The system shall offer an easily usable interface by people of various ages.	Y
The interface of the chatbot shall be easy to use.	Y
The system shall offer an engaging experience for cultural interactions.	Y
The conversational responses generated by the system shall remain clear and readable	Y
Accuracy Requirements	
The system must deliver accurate information about the Palestinian culture.	Y
The system must not generate false or irrelevant answers.	Y

The chatbot avatars must maintain consistency in all interactions.	Y
Scalability Requirements	
The system must enable the addition of new cultural information and artifacts in the future.	Y
The AI Wiki must enable the continuing growth of the knowledge base.	Y
The system architecture must fit the addition of new chatbot characters in subsequent releases.	Y
Security Requirements	
The system will limit access to the admin management interface.	Y
The system will protect the AI Wiki database against any unauthorized alterations.	Y
The system will ensure secure storage of cultural information and system information.	Y
Maintainability Requirements	
The system will ease the simple update of stories, objects, and cultural information.	Y
The system will provide for the maintenance and adjustment of the chatbot's conversation patterns.	Y
The system will enable effective management of cultural content by the administrators.	Y
Compatibility Requirements	
The application should be compatible with supported desktop operating systems.	Y

The system should support the capability of integrating with 3D virtual environment technology.	Y
The system should support the interaction and display of content using Arabic	Y
Cultural Experience Requirements	
The system should continue the Palestinian cultural identity through its interface design and conversation.	Y
The system should offer an immersive storytelling experience.	Y
The system should sustain authentic Palestinian expressions and culture in the chatbot conversations.	Y
The system should provide an enriching education experience on the Palestinian culture.	Y

Table 5.1 2

5.2 UI Testing

UI testing was conducted to check whether the designed interfaces of **Palestinian Cultural Shop** cover the main features and specifications mentioned in the project. The test focuses on the visual and interaction elements of the application, such as the shop environment, navigation controls, clickable heritage objects, 3D object viewer, and chatbot interface

Feature	UI Testing Result	Test Status
Welcome screen	The design includes a welcome interface that introduces the user to the application and provides a clear login button.	Y
Login screen	The UI includes username and password fields, a login button, and additional account options.	Y
Virtual Palestinian heritage shop	The design shows a complete Palestinian heritage shop with traditional sections, objects, and cultural identity.	Y
Shop sections	The shop is divided into clear sections such as pottery, embroidery, traditional tools, old photos, reception, and featured objects.	Y
Navigation controls	Movement buttons are displayed clearly to help the user move inside the virtual shop.	Y
Clickable heritage objects	Heritage objects such as pottery, embroidery pieces, coffee pot, books, and old tools are visually clear and can be clicked.	Y
Object interaction instruction	The UI includes visual instructions such as “Tap on the item” to guide the user.	Y
3D object viewer And object rotation	The design includes a screen where the selected object appears in the center while the background is blurred and can be moved and rotated from different angles.	Y
Grandfather chatbot	The grandfather chatbot is represented by an icon and a chat interface, allowing users to ask questions about tangible heritage objects.	Y
Grandmother chatbot	The grandmother character is shown in the embroidery section, supporting the storytelling and intangible heritage part of the project.	Y
Chatbot conversation screen	The chatbot screen shows a conversation between the user and the grandfather, including a welcome message and a text input field.	Y
Reception area	The reception area includes the grandfather, a wooden table, coffee pot, cups, calculator, and Palestine map, which supports the idea of a traditional shop entrance.	Y
Mobile interface	The screens are designed vertically to fit a mobile application layout.	Y
Palestine identity	The design includes a large Palestine map, traditional colors, embroidery patterns, stone walls, wooden shelves, lanterns, and heritage objects.	Y
User guidance	During the first use, the UI includes arrows, icons, labels, and short instructions to guide the user while exploring.	Y

Visual consistency	The design uses consistent warm colors, traditional materials, and heritage elements across all screens.	Y
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Table 5.1 3

5.3 UI Testing Summary

The UI testing shows that the designed interfaces cover the main visual and interaction requirements of the application. The design supports the idea of a 3D Palestinian heritage shop, includes clear navigation controls, presents clickable heritage objects, provides a 3D object viewer, and includes chatbot interaction with the grandfather and grandmother characters. Therefore, the current UI design matches the main project concept and supports the user experience described in the project requirements

Chapter 6: Conclusion

To conclude, Rawasi: Palestinian Cultural Shop is an example of an advanced approach towards cultural preservation and promotion through interactive technologies. It includes elements like 3D virtual environment, cultural objects, and artificial intelligence chatbot characters, thus providing users with an exciting educational experience. With the help of such features as exploration of a virtual heritage shop, interaction with cultural objects and chatting with the grandfather and Grandmother bots, the presented application enables its users to make an interactive journey in discovering the culture instead of studying it by just reading the available materials.

It should be admitted that Rawasi meets all the requirements for the project as it provides a possibility to present the culture in a convenient, interesting and easily comprehensible form. In contrast to ordinary websites dedicated to cultures, the project focuses on interactive features, gaming aspects and cultural identity, which is achieved with the help of AI Wiki, chatbot agents, 3D objects and admin management.

Additionally, the test results show that the key functionalities and non-functionalities have been met, such as user navigation, interaction with objects, chatbot communications, cultural facts, usability, performance, scalability, and security. At the same time, the UI tests prove that the design of the application shows the uniqueness of Palestinian culture through the use of traditional colours, objects, maps, patterns, and cultural symbols.

Overall, Rawasi is a good example of using technology to preserve culture through providing the heritage in a modern and interactive form. The project has the potential to raise cultural awareness, reinforce Palestinian identity, and allow the users to explore the heritage. In addition, there are possible areas for future development, such as increasing the number of artifacts, AI Wiki entries, chatbot capabilities, platform availability, and making the 3D experience more realistic.

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